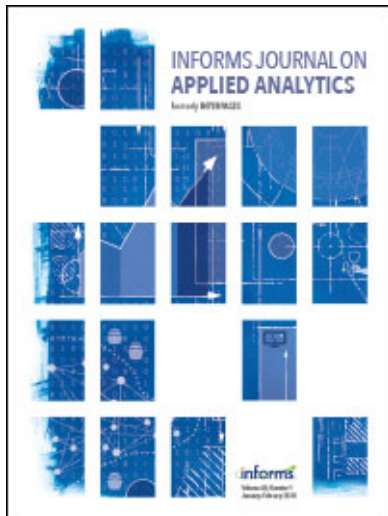




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A Proposal for Redesign of the FedEx Cup Playoff Series on the PGA TOUR

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We propose a new design for the FedEx Cup playoff series on the PGA TOUR, and especially for its final event, the TOUR Championship. Playing the TOUR Championship as a strongly seeded match-play event, as in our proposal, provides a good chance of an exciting climax to the professional golf season. We validate our proposed design by estimating the probability that a player who has a particular rank entering the TOUR Championship finishes in various positions in the FedEx Cup standings. The choice of match play in our design is motivated by the belief held by various stakeholders in professional golf that match play is closer to the spirit of a playoff event than stroke play. Moreover, our strongly seeded match-play design provides several advantages, including the following: all players have a reasonable opportunity to win at the TOUR Championship; all players are rewarded for consistently strong performance during the season; and the participation of marquee name players is guaranteed during network television coverage late in the event. We discuss how the proposed design achieves the various objectives outlined by PGA TOUR Commissioner Tim Finchem about possible redesign of the FedEx Cup playoff series.

Key words: sports application; professional golf administration; probabilistic analysis; sensitivity analysis.

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Professional golf, a worldwide major sport, comprises six major tours, which are organized by geographical region: the PGA TOUR (United States), the European Tour, the Asian Tour, the Japan Tour, the PGA Tour of Australasia, and the Sunshine Tour (South Africa). Statistics for the PGA TOUR indicate the commercial importance of professional golf. In the 2009 season, the PGA TOUR scheduled 47 events with about \$275 million in prize money. In a typical year, the leading money winner earns about \$10 million in prize money. The winner of the end-of-season playoff series receives a similar amount. The television contracts of the PGA TOUR for the six-year period 2007–2012 are worth approximately \$2.6 billion. The PGA TOUR is proud of its historical record of charitable donations, which total about \$1.2 billion; this amount is larger than the charitable donations from all other US major sports combined. The PGA TOUR is also

instrumental in providing visibility and leadership to the sport of golf, which had an estimated economic value of \$195 billion and provided two million jobs to the US economy in 2005 (PGA TOUR 2008a).

The PGA TOUR initiated the FedEx Cup playoff series in 2007. A major objective was to bring the lengthy professional golf season to an exciting climax, similar to that enjoyed by other major sports in the United States and worldwide.

The FedEx Cup system consists of a season-long points competition and an end-of-season playoff series. For the officially sanctioned events throughout the regular season, players receive FedEx Cup points based on their finishing positions in each tournament. Entry into the playoff series is based on the total number of points accumulated during the regular season.

The FedEx Cup playoff series consists of four events. The first event, The Barclays Championship,

starts with the 125 players who have earned the most points during the regular season. However, the number of players who are invited to progress to the next event is reduced each week, until 30 players compete in the final event in the series, the TOUR Championship. Each of the four playoff-series events is a 72-hole stroke-play tournament, at which points are awarded. Using their current points, the players are ranked to determine who is invited to play in the next event.

At each of the four playoff-series events, prize money is awarded based on a player's finishing position. In addition, the FedEx Cup offers a bonus pool of \$35 million in prize money to the leading players in the points standings following the TOUR Championship. For example, the winner receives \$10 million and the stylish FedEx Cup trophy, while the fifth-placed player receives \$1 million; all 125 players receive some prize money.

The format of the FedEx Cup playoff series can be criticized for various reasons. The most serious of these is that a strong performance in the TOUR Championship is not always necessary to win the FedEx Cup. As a result, the climax to the golf season might not be exciting. In this paper, we propose an improved design for the FedEx Cup playoff series that avoids most of the criticisms associated with the current format.

This paper is organized as follows. The *Outcomes and Commentary* section discusses the outcomes from the 2007–2009 FedEx Cup playoff series and some resulting media commentary, and identifies several key objectives for a possible redesign of the FedEx Cup. The *Related Literature* section discusses the literature on sports statistics. In *Alternative Playoff Designs*, we provide an overview of several different possible playoff-series designs, and their limitations. The *Proposed Design* section contains a summary of our suggested design. In *Analysis of the Proposed Design*, we perform a probabilistic analysis of our proposed design and present numerical results for typical outcomes that would result. *Advantages of the Proposed Design* summarizes the ways in which our proposed design meets the objectives discussed in the *Outcomes and Commentary* section. In *Extensions*, we identify several extensions and variations of our design that could be used to add further excitement to the event. Finally, we complete the paper with a *Conclusions* section.

Outcomes and Commentary

Unfortunately, the outcomes of the FedEx Cup playoff series were unexciting in both 2007 and 2008. In 2007, Tiger Woods entered the playoff series with a substantial points lead, finished second at the second playoff event, and won the third playoff event. Consequently, the FedEx Cup winner was decided before the fourth event started. As a result of this anticlimactic outcome, the point spreads resulting from the 2008 regular season were reduced, and those resulting from the 2008 playoff series were increased, to bring more players into contention during the playoffs.

In 2008, Vijay Singh won the first two playoff-series events. Therefore, he had only to show up and play at the TOUR Championship to win the FedEx Cup. Although he played poorly in that event, he still took home the \$10 million winner's check and the trophy, thus providing another anticlimax. In addition, a combination of the adjustments in the points system and some mediocre play eliminated Padraig Harrington from the TOUR Championship. Because Harrington was arguably the best active player in the world in the absence through injury of Tiger Woods, this was an unfortunate outcome for spectators. An additional concern in both 2007 and 2008 was a lack of consistent participation by some leading players throughout the playoffs.

The media published various negative comments about the FedEx Cup format. For example, the following commentary about the first-year experience with the FedEx Cup ran: "Ho-hum. The FedEx Cup is over. Finally. It made Tiger Woods \$10 million richer (in escrow), but did it make the PGA any richer? Probably not. . . . The FedEx Cup seemed no different than a regular tour event, even lesser in stature than the World Golf Championship events. . . . The scoring system was a mystery and left only 3 players in contention for the final tournament. Why not start the final 30 all from scratch for the final tourney?" (The Golf Blog 2007, p. 1).

Golf legend Jack Nicklaus offered some negative comments about the second year of the FedEx Cup. "Did they play that this year?" asked Nicklaus. "I didn't watch a second of it. They still have some tweaking to do" he continued. It has a tendency to take away from the rest of the tour events. It's always, "He's got X-amount of FedEx Cup points." A guy

doesn't give a rat about FedEx points, he cares about how he's playing" (*Los Angeles Times* 2008, p. 1).

Contributing to the need for a revised format for the FedEx Cup are concerns about potential competition. In 2008, the European Golf Tour announced plans to launch a similar end-of-season event with substantial bonus money at the 2009 Dubai World Championships. Commentary on Golf Fan House (2008) suggests, "Obviously, this spells doom for a FedEx Cup system that hasn't necessarily taken off like they'd hoped. The Cup will have to be revamped before next season so someone like Singh can't win the thing by the third tournament.... The PGA TOUR better watch out. As much as we all love Tiger, when he comes back, with his interests in Dubai close at heart, you're telling me he won't be a part of this thing?"

The PGA TOUR clearly recognizes these concerns. A blog by Commissioner Tim Finchem (PGA TOUR 2008b) includes three key objectives for future improvement.

OBJECTIVE 1. The playoffs should build towards a climactic finish, meaning the FedEx Cup should be decided at the TOUR Championship.

OBJECTIVE 2. The FedEx Cup competition should culminate in a dramatic way, meaning that a reasonable number of players should be in strong contention to win it at the TOUR Championship.

OBJECTIVE 3. The TOUR Championship field should consist of the top performers in the regular season in addition to players who have excelled in the playoffs.

In addition, the players have a fourth objective, which concerns fairness of the playoffs and the distribution of prize money in relation to performance.

OBJECTIVE 4. A player's performance during the regular season should impact his finishing position in the FedEx Cup and his resulting prize money.

To achieve the above objectives, the points system was adjusted for the 2009 FedEx Cup series (Golf.com 2008). The main adjustment was that the season-long accumulated points would not be reset until before the TOUR Championship. This would give every player at the TOUR Championship at least

a mathematical chance of winning the FedEx Cup, although only the top-five-ranked players *controlled their own destiny*. That is, only those players could win the FedEx Cup by winning the TOUR Championship, irrespective of the performance of all other players. Phil Mickelson won the TOUR Championship in dramatic style. However, Tiger Woods won the FedEx Cup on the strength of a season in which he won six tournaments including the third playoff series event, the BMW Championship.

The points system used in 2009 also failed to provide adequate incentives for consistent participation by the players. For example, Geoff Ogilvy and Brian Gay entered the playoffs ranked eighth and ninth, respectively. They could have missed all the first three playoff-series events and still entered the TOUR Championship ranked 27th and 30th, respectively. As another example, Heath Slocum won the first playoff-series event. Had he missed the second and third playoff-series events, he would have entered the TOUR Championship ranked fifth, and still in control of his own destiny.

Although the FedEx Cup outcome was closer in 2009 than in the two previous years, it failed to generate as much public enthusiasm as hoped. Moreover, several influential golf writers believe that match play would be a better alternative, provided that some issues can be resolved. Van Sickle (2009) writes, "A real playoff system would probably involve match play, but that isn't realistic because it would defeat the event's primary mission—to get the best golfers playing more often. As we've seen in the World Match Play event in Tucson, the top players often get bounced, sometimes in the early rounds." David Dusek echoes this viewpoint, "...it's too bad the TOUR Championship can't be a match play event. The powers that be are just too afraid of losing stars in early rounds" (Golf.com 2009a). Similarly, Farrell Evans comments, "...it will never be perfect unless there is a match play component" (Golf.com 2009b). In this paper, we propose a match-play format for the TOUR Championship that resolves this concern about early elimination of the top players.

Related Literature

Schwertman et al. (1991) consider three statistical models that use rankings to predict the outcomes of

head-to-head matches in National Collegiate Athletic Association (NCAA) men's basketball. Let r_i denote the ranking of competitor i , where, for example, $r_i = 1$ if i is the top-ranked competitor. The first model uses $r_j/(r_i + r_j)$ as the probability of competitor i beating competitor j , based on the belief that teams with a given difference in ranking are closer in ability when both are lower in the ranking than when both are higher. The second model uses $0.5 + (r_j - r_i)\Delta$ as the probability, where Δ is a nonnegative constant that is estimated by linear regression. The third model is similar to the second one; however, it assumes a normal rather than uniform distribution of abilities. The third model provides the best fit with empirical data.

Schwertman et al. (1996) extend the above analysis to consider 11 probability models, including those based on both ordinary and logistic regression. To predict the outcome of head-to-head matches, one of the ordinary regression models provides the best fit with empirical data. However, a logistic regression model is more accurate at predicting the winner of a single elimination tournament. Smith and Schwertman (1999) use ranking to predict margin of victory in NCAA men's basketball. They achieve accurate predictions using a quadratic and linear model of rankings. Carlin (1996) incorporates publicly available data besides ranking (e.g., point spreads) into a model for predicting the outcomes of NCAA men's basketball games. However, such information is typically not available for match-play golf. Caudill and Godwin (2002) develop a more general model that allows for the possibility of heterogeneous skewness, thereby allowing skewness to vary from one match to another. They model skewness as a linear and quadratic function of the higher of the two rankings. The results of statistical tests on data from NCAA men's basketball reject previous models in favor of the model with heterogeneous skewness.

Boulier and Stekler (1999) study NCAA basketball and professional tennis, for both men and women, using probit models. These models relate the probability of occurrence of a discrete zero-one event to several explanatory variables. The study concludes that although knowledge of the rankings of teams is useful for forecasting, including the difference in rankings as an explanatory variable improves the accuracy of predictions. Noting that the probit models of Boulier and

Stekler (1999) are not calibrated to maximize the number of correct predictions, Caudill (2003) develops a maximum-score estimator that does so. As a method for classifying games into "win" or "lose," the proposed method is superior to probit models. However, it does not provide the probability estimates that our study requires.

Kaplan and Gartksa (2001) consider a single elimination tournament in which the estimated probability of any team beating any other team is known. They derive a prediction strategy, which is based on dynamic programming, to maximize a bettor's score within an office pool.

Kvam and Sokol (2006) consider the problem of how to rank teams using limited publicly available data, such as online scoreboards. They use a logistic regression model to estimate the transition probabilities of a Markov chain. Teams are then ranked based on their steady-state probabilities from the Markov chain. The model predicts the results of head-to-head matches better than other models based on previous matches within the same round robin schedule. However, in our study, the round robin data are not available.

Alternative Playoff Designs

Many minor adjustments could be made to the FedEx Cup points system, while retaining the series of four stroke-play events. However, finding the right adjustments may be difficult. Consensus opinion seems to be that the points system used in 2007 offered too little opportunity for players to catch up; hence, too few players came into the playoff series with a chance to win. The revised points system used in 2008 was designed to resolve these problems. However, as discussed above, it still resulted in an anticlimactic ending, as well as some additional problems. The revised points system used in 2009 only partly resolved these issues. For example, leading players Jim Furyk and Jerry Kelly expressed the view that it gave too much weight to performance during the playoff events, relative to performance in the regular season (Golf Channel 2009).

A more radical change, advocated by several writers and Golf Channel analysts, would make the

TOUR Championship a match-play event. Among the advantages are that match play is the usual play-off format in most other sports, is often exciting and unpredictable, and may create opportunities for more players to win the FedEx Cup. The match-play alternative could be achieved in at least two ways, each with its own advantages and disadvantages.

First, the current FedEx Cup system, in which the winner is the player with the most total points, could be continued. However, if the match-play event awards relatively too few points, then the FedEx Cup may be effectively over before that event starts. Conversely, if the match-play event awards relatively too many points, then this one event may override the effect of consistent play throughout the regular season. This seems unfair to players who have performed consistently well throughout the season. The latter concern is particularly relevant because of the unpredictable nature of match-play events. However, achieving the right balance is problematic because it is difficult to predict the distribution of total point scores entering the match-play event. A further concern is that the winner of the TOUR Championship would likely not be the winner of the FedEx Cup based on total points. Indeed, this occurred in 2008 and 2009 under the playoff systems then in place. This is in contrast to the playoff outcomes of all other major US sports.

Second, the point totals entering the TOUR Championship could be used to establish a seeding for the match-play event, similar to that used for the World Golf Championship (WGC)-Accenture match-play event, and then not considered further. The FedEx Cup results would be based solely on the TOUR Championship results. However, because of the unpredictability of match-play events, this format also seems unfair to players who have performed consistently well throughout the season. That is, this design makes it too easy for a player with a mediocre performance during the regular season to win the FedEx Cup. On this basis, it seems that the seeding for the TOUR Championship is not strong enough.

For these reasons, we propose a complete redesign of the FedEx Cup playoff series format and the TOUR Championship, as we discuss in the next section.

Proposed Design

Our proposed design for the FedEx Cup consists of an overall format and a new design for the TOUR Championship.

FedEx Cup Format

1. Players will accumulate FedEx Cup points both during the regular season and during the first three events of the playoff series, as they do at present.
2. The top 28 players in the point standings after the first three playoff events will be invited to play in the TOUR Championship; 30 players are currently invited.
3. The rankings of the top 28 players will be used to seed at the TOUR Championship. However, after seeding, the point totals will not be used again.
4. The TOUR Championship will be played using a strongly seeded match-play format, which we describe below.
5. The finishing order of the players at the TOUR Championship, including ties, will determine the finishing order of the top 28 places in the FedEx Cup.

Redesign of the TOUR Championship

The TOUR Championship will admit the 28 players with the highest point totals accumulated over the season and the first three playoff-series events. The format will be a four-day, eight-round, 18-hole match-play event, from Thursday through Sunday (see Table 1). This results in four matches in the morning and four in the afternoon on Thursday, Friday, and Saturday, two matches (semifinals) on Sunday morning, and two matches (final and playoff for third and fourth place) on Sunday afternoon. Therefore, the total number of matches is 28.

For each of the four matches in the morning and afternoon of Thursday, Friday, and Saturday, the lowest-ranked player among the winners from the previous round is matched with the highest-ranked of the players just entering the tournament, the second-lowest-ranked player of the previous winners is matched with the second-highest-ranked of the players just entering, and so on. In the semifinals on Sunday morning, the highest- and lowest-ranked of the four winners from Saturday afternoon play each other. This process of creating matches by assigning the highest-ranked player to play the lowest-ranked

Session	Match format
Thursday AM	Players ranked 21–28 in match play (rank 21 vs. rank 28, rank 22 vs. rank 27, and so on)
Thursday PM	The four winners from Thursday AM in match play against the players ranked 17–20 (highest ranked vs. lowest ranked, and so on)
Friday AM	The four winners from Thursday PM in match play against the players ranked 13–16 (highest ranked vs. lowest ranked, and so on)
Friday PM	The four winners from Friday AM in match play against the players ranked 9–12 (highest ranked vs. lowest ranked, and so on)
Saturday AM	The four winners from Friday PM in match play against the players ranked 5–8 (highest ranked vs. lowest ranked, and so on)
Saturday PM	The four winners from Saturday AM in match play against the players ranked 1–4 (highest ranked vs. lowest ranked, and so on)
Sunday AM	Two semifinal matches among the four winners from Saturday PM (highest ranked vs. lowest ranked, and so on)
Sunday PM	Championship and 3rd/4th place matches

Table 1: Our redesign of the TOUR Championship includes the individual matches, as shown in the table.

player provides a reward for being the higher-ranked player.

Analysis of the Proposed Design

In this section, we introduce a stochastic model to estimate the probabilities of the various players winning under our proposed redesign of the TOUR Championship, based on the ranking of the players at its start. Our aim is to evaluate whether the proposed design gives all the players a reasonable chance to win and also appropriately rewards players who enter the TOUR Championship with a higher rank based on consistently good performance earlier in the season.

We first develop the model for our probabilistic analysis. We then present some numerical results that indicate the probabilities of the players progressing to different stages of the tournament based on their rank.

Probabilistic Analysis

Let the players be indexed $1, \dots, 28$ according to their rank, so that player 1 is the highest ranked, player 2

is the second highest, and so on, with player 28 being the lowest ranked. We refer to the sets of matches described in the eight rows of Table 1 as rounds $1, \dots, 8$, respectively.

Let q_{ij} denote the probability that player i beats player j , where $q_{ij} + q_{ji} = 1$ for $i, j = 1, \dots, 28$, and $i \neq j$. A player's chance of progressing to the next round depends on whom he plays in the current round, which in turn depends on the other players who have progressed to the current round according to our proposed design of the TOUR Championship. Therefore, we define $p_{j_1, j_2, j_3, j_4}(r)$ to be the probability that players j_1, j_2, j_3 , and j_4 progress from round r to round $r + 1$ for $r = 1, \dots, 6$ and $28 \geq j_1 > j_2 > j_3 > j_4 \geq 25 - 4r$. Analogously, $p_{j_1, j_2, j_3, j_4}(7)$ for $28 \geq j_1 > j_2 \geq 1$ and $28 \geq j_3 > j_4 \geq 1$ is the probability that players j_1 and j_2 win the two semifinals, and that players j_3 and j_4 lose the two semifinals. Further, $p_{j_1, j_2, j_3, j_4}(8)$ is the probability that players j_1, j_2, j_3 , and j_4 , respectively, occupy the first four positions at the end of the competition, where $1 \leq j_k \leq 28$ for $k = 1, 2, 3, 4$. For the initial conditions, we set

$$p_{28, 27, 26, 25}(0) = 1 \quad (1)$$

and $p_{j_1, j_2, j_3, j_4}(0) = 0$ for all other combinations of j_1, j_2, j_3 , and j_4 .

We next outline how the probabilities $p_{k_1, k_2, k_3, k_4}(r)$ are evaluated for the relevant values of k_1, k_2, k_3, k_4 , and r . The appendix includes a complete listing of the equations.

Through the evolution of the tournament, the probabilities $p_{k_1, k_2, k_3, k_4}(r)$ are evaluated recursively from the probabilities of progressing into round r for $r = 1, \dots, 6$ and $28 \geq k_1 > k_2 \geq k_3 > k_4 \geq 25 - 4r$. Let $i_1 = 25 - 4r$, $i_2 = 26 - 4r$, $i_3 = 27 - 4r$, and $i_4 = 28 - 4r$ be the four highest-ranked players competing in round r ; let j_1, j_2, j_3 , and j_4 be the four lowest-ranked players in round r , where $28 \geq j_1 > j_2 > j_3 > j_4 \geq 29 - 4r$. Thus, players i_1 and j_1 , i_2 and j_2 , i_3 and j_3 , and i_4 and j_4 play the matches in round r . Although $i_1, \dots, i_4, j_1, \dots, j_4$ are dependent on the round r , for simplicity we suppress this notation because the round being considered is clear from the context. We assume that match outcomes are probabilistically independent of each other and of the round in which the match is played. There are 16 possible outcomes for the four

players progressing into the next round from the matches involving the pairings (i_1, j_1) , (i_2, j_2) , (i_3, j_3) , and (i_4, j_4) , and each has its own probability equation.

Similarly, we evaluate the probabilities p_{j_1, j_2, j_3, j_4} (7) recursively from the probabilities of progressing from round 6, where j_1 and j_2 are the two semifinal winners, for distinct j_1, j_2, j_3 , and j_4 , and for $j_1 > j_2$ and $j_3 > j_4$. Further, we evaluate the probability p_{j_1, j_2, j_3, j_4} (8) of players j_1, j_2, j_3 , and j_4 occupying the first four positions, based on the probabilities of results achieved in the semifinals.

From these probabilities, the probability of player j winning the TOUR Championship is given by

$$\sum_{j_2, j_3, j_4} p_{j, j_2, j_3, j_4} \quad (8), \quad (2)$$

where the summation is over values of j_2, j_3 , and j_4 , which are distinct from each other and from j . We are also able to compute the probability of player j finishing in any position in the TOUR Championship (see the appendix).

Numerical Evaluation

Our numerical evaluation is aimed at assessing how our proposed redesign performs in terms of the probabilities of the different ranked players occupying the various finishing positions. To apply the probabilistic analysis we discuss in the *Probabilistic Analysis* subsection, we describe a probability model that defines the value of $q_{i,j}$ for an individual match between players ranked i and j . As above, we assume that the players are indexed $1, 2, \dots$, so that player 1 is the highest ranked, player 2 is the second-highest ranked, and so on.

Limited data are available to determine a probability model for individual matches. We use data from the annual WGC-Accenture match-play championships during 2004–2010. This produces 64 matches for each of the seven years, resulting in 448 matches.

Based on the results of Schwertman et al. (1991, 1996), as discussed in the *Related Literature* section, the model with $q_{i,j} = j/(i+j)$ as the probability of player i beating player j in an individual match has some advantages. First, the model does not require any parameters to be chosen, unlike several alternative models. This is an important consideration in view of our limited data. Second, Schwertman et al. (1991, 1996) show that the model's fit with the NCAA basketball data is reasonable, relative to that

of the other methods considered. However, a possible drawback is that the probabilities are too large for matches involving highly ranked players. For example, the probability of 2/3 for the highest-ranked player beating the second-highest might be regarded as unrealistically high.

We now discuss how the potential drawback mentioned above can be overcome. We first define an *adjusted rank* $r'_k = r_k + \delta$ for each player k , where $\delta \geq 0$ is a parameter. Using the adjusted ranks, we obtain the model

$$q_{i,j} = \frac{r'_j}{r'_i + r'_j} = \frac{r_j + \delta}{r_i + r_j + 2\delta}. \quad (3)$$

As δ becomes larger, the drawback of the probabilities being too large for matches involving two highly ranked players disappears.

The value of the parameter δ is chosen to minimize the sum, over all players k , of the absolute deviations of the observed number of wins achieved by player k among the 448 WHC-Accenture matches, minus the expected number of wins for player k , which we calculate using Equation (3). The value $\delta = 33.5$ provides the best fit.

Therefore, our first study uses $\delta = 33.5$ (see Table 2). The second and third studies are designed to frame these results by studying smaller and larger advantages, respectively, for the higher-ranked player. The aim is to ensure that even if our above estimate (i.e., $\delta = 33.5$) from the WHC-Accenture data does not reflect accurate probabilities, the results are still reasonable. In the second study (see Table 3), we assume that $\delta = 67.0$ (double the value in Table 2); in the third study (see Table 4), we assume that $\delta = 16.75$ (half the value in Table 2).

The results of applying the probabilistic analysis as defined by Equations (1)–(22) are displayed in Tables 2–4, which show the percentage probability with which a player of a particular rank (R) entering the TOUR Championship finishes in a particular position (F). For example, in Table 2, a player who is ranked seventh (row R7) wins the TOUR Championship and the FedEx Cup (column F1) 5.9 percent of the time. The notation “FT” in a column indicates players finishing in a tie; for example, FT5 indicates a tie for fifth through eighth places for the four players who are eliminated on Saturday PM.

Rank	F1	F2	F3	F4	FT5	FT9	FT13	FT17	FT21	FT25
R1	16.5	15.2	13.7	12.7	41.8	0.0	0.0	0.0	0.0	0.0
R2	15.3	14.4	13.1	12.3	44.9	0.0	0.0	0.0	0.0	0.0
R3	14.1	13.3	13.3	12.4	47.0	0.0	0.0	0.0	0.0	0.0
R4	13.2	12.7	13.0	12.3	48.8	0.0	0.0	0.0	0.0	0.0
R5	7.1	7.1	7.2	7.1	29.1	42.5	0.0	0.0	0.0	0.0
R6	6.4	6.6	6.7	6.7	28.3	45.3	0.0	0.0	0.0	0.0
R7	5.9	6.1	6.4	6.5	27.8	47.2	0.0	0.0	0.0	0.0
R8	5.5	5.7	6.1	6.3	27.4	48.9	0.0	0.0	0.0	0.0
R9	2.9	3.1	3.3	3.5	15.2	28.7	43.3	0.0	0.0	0.0
R10	2.6	2.9	3.1	3.3	14.5	28.0	45.7	0.0	0.0	0.0
R11	2.4	2.6	2.9	3.2	13.9	27.5	47.5	0.0	0.0	0.0
R12	2.2	2.5	2.7	3.0	13.4	27.2	49.0	0.0	0.0	0.0
R13	1.1	1.3	1.4	1.6	7.3	14.8	28.2	44.2	0.0	0.0
R14	1.0	1.2	1.3	1.5	6.9	14.3	27.7	46.1	0.0	0.0
R15	0.9	1.1	1.2	1.5	6.5	13.8	27.3	47.6	0.0	0.0
R16	0.8	1.0	1.1	1.4	6.2	13.4	27.0	49.1	0.0	0.0
R17	0.4	0.5	0.6	0.7	3.2	7.1	14.4	27.5	45.5	0.0
R18	0.4	0.5	0.5	0.7	3.1	6.8	14.1	27.3	46.7	0.0
R19	0.3	0.4	0.5	0.6	2.9	6.5	13.7	27.1	47.9	0.0
R20	0.3	0.4	0.5	0.6	2.7	6.2	13.3	26.8	49.2	0.0
R21	0.2	0.2	0.2	0.3	1.4	3.2	6.9	14.0	26.8	47.0
R22	0.1	0.2	0.2	0.3	1.3	3.0	6.6	13.7	26.7	47.8
R23	0.1	0.2	0.2	0.3	1.2	2.9	6.4	13.4	26.6	48.7
R24	0.1	0.2	0.2	0.2	1.2	2.8	6.2	13.1	26.5	49.6
R25	0.1	0.1	0.2	0.2	1.1	2.6	6.0	12.9	26.3	50.4
R26	0.1	0.1	0.2	0.2	1.0	2.5	5.8	12.6	26.1	51.3
R27	0.1	0.1	0.1	0.2	1.0	2.4	5.6	12.3	26.0	52.2
R28	0.1	0.1	0.1	0.2	0.9	2.3	5.4	12.0	25.8	53.0

Table 2: The table shows the percentage probability of FedEx Cup outcomes: $\delta = 33.5$.

From the results in Table 2, where $\delta = 33.5$, the FedEx Cup is won by the top-ranked player 16.5 percent of the time. One of the top-four-ranked players wins 59.1 percent of the time. We believe that this result is appropriate because having a marquee name champion clearly adds credibility and prestige to the event. Also, the players ranked 5th through 8th collectively win 24.9 percent of the time, and the players ranked 9th through 12th collectively win 10.1 percent of the time, both of which suggest that winning is a realistic possibility for others and add to the excitement and uncertainty of the event. Players ranked 13th through 16th collectively win 3.8 percent of the time, which is also a realistic possibility. Finally, even the 28th-ranked player wins occasionally.

The results in Table 3, where $\delta = 67.0$ percent, differ only slightly and provide confirmation of the reasonableness of the design. The top-ranked player wins

Rank	F1	F2	F3	F4	FT5	FT9	FT13	FT17	FT21	FT25
R1	14.7	14.1	13.3	12.7	45.3	0.0	0.0	0.0	0.0	0.0
R2	14.1	13.6	12.8	12.4	47.1	0.0	0.0	0.0	0.0	0.0
R3	13.4	13.0	12.9	12.4	48.3	0.0	0.0	0.0	0.0	0.0
R4	13.0	12.6	12.7	12.3	49.3	0.0	0.0	0.0	0.0	0.0
R5	6.8	6.7	6.8	6.7	27.4	45.6	0.0	0.0	0.0	0.0
R6	6.4	6.4	6.5	6.5	26.8	47.2	0.0	0.0	0.0	0.0
R7	6.1	6.2	6.4	6.4	26.5	48.4	0.0	0.0	0.0	0.0
R8	5.9	6.0	6.2	6.3	26.3	49.4	0.0	0.0	0.0	0.0
R9	3.0	3.2	3.3	3.4	14.0	27.2	45.9	0.0	0.0	0.0
R10	2.9	3.0	3.1	3.3	13.6	26.7	47.4	0.0	0.0	0.0
R11	2.7	2.9	3.0	3.2	13.3	26.5	48.5	0.0	0.0	0.0
R12	2.6	2.8	2.9	3.1	13.0	26.2	49.4	0.0	0.0	0.0
R13	1.3	1.4	1.5	1.6	6.9	13.9	26.9	46.5	0.0	0.0
R14	1.2	1.4	1.4	1.6	6.6	13.6	26.6	47.6	0.0	0.0
R15	1.2	1.3	1.4	1.5	6.4	13.3	26.4	48.5	0.0	0.0
R16	1.1	1.2	1.3	1.5	6.2	13.0	26.2	49.4	0.0	0.0
R17	0.6	0.6	0.7	0.8	3.2	6.8	13.7	26.6	47.2	0.0
R18	0.5	0.6	0.6	0.7	3.1	6.6	13.4	26.4	48.0	0.0
R19	0.5	0.6	0.6	0.7	3.0	6.4	13.2	26.3	48.7	0.0
R20	0.5	0.5	0.6	0.7	2.9	6.3	13.0	26.1	49.5	0.0
R21	0.2	0.3	0.3	0.3	1.5	3.2	6.6	13.4	26.1	48.1
R22	0.2	0.3	0.3	0.3	1.4	3.1	6.5	13.2	26.1	48.6
R23	0.2	0.2	0.3	0.3	1.4	3.0	6.4	13.1	26.0	49.1
R24	0.2	0.2	0.3	0.3	1.3	2.9	6.2	12.9	25.9	49.7
R25	0.2	0.2	0.2	0.2	1.3	2.8	6.1	12.8	25.8	50.3
R26	0.2	0.2	0.2	0.2	1.2	2.8	6.0	12.6	25.7	50.8
R27	0.2	0.2	0.2	0.2	1.2	2.7	5.9	12.4	25.6	51.4
R28	0.2	0.2	0.2	0.2	1.2	2.6	5.7	12.3	25.5	51.9

Table 3: The table shows the percentage probability of FedEx Cup outcomes: $\delta = 67.0$.

14.7 percent of the time, and the top-four-ranked players collectively win 55.2 percent of the time. Lower-ranked players benefit slightly under this larger value of δ . For example, players ranked 13th through 16th collectively win 4.8 percent of the time.

Similarly, the results in Table 4, where $\delta = 16.75$, provide additional confirmation. The top-ranked player wins 19.5 percent of the time, and the top-four-ranked players collectively win 64.5 percent of the time. Lower-ranked players perform slightly worse than above. For example, players ranked 13th through 16th collectively win 2.6 percent of the time.

The results in Tables 3 and 4 suggest that the outcome of the proposed design for the TOUR Championship is not sensitive to the value of δ . More precisely, the degree of superiority that a higher-ranked player has over a lower-ranked player does not make a substantial difference to the overall outcome.

Rank	F1	F2	F3	F4	FT5	FT9	FT13	FT17	FT21	FT25
R1	19.5	17.0	14.3	12.4	36.8	0.0	0.0	0.0	0.0	0.0
R2	17.0	15.6	13.6	12.3	41.5	0.0	0.0	0.0	0.0	0.0
R3	14.8	13.9	13.9	12.5	44.9	0.0	0.0	0.0	0.0	0.0
R4	13.2	12.9	13.4	12.4	48.1	0.0	0.0	0.0	0.0	0.0
R5	7.2	7.5	7.7	7.6	31.4	38.6	0.0	0.0	0.0	0.0
R6	6.2	6.6	6.9	7.1	30.4	42.6	0.0	0.0	0.0	0.0
R7	5.4	5.9	6.5	6.8	29.8	45.6	0.0	0.0	0.0	0.0
R8	4.7	5.3	6.0	6.4	29.2	48.3	0.0	0.0	0.0	0.0
R9	2.5	2.9	3.3	3.7	16.9	30.6	40.1	0.0	0.0	0.0
R10	2.1	2.6	2.9	3.4	15.8	29.7	43.5	0.0	0.0	0.0
R11	1.8	2.3	2.6	3.2	14.8	29.1	46.2	0.0	0.0	0.0
R12	1.6	2.0	2.4	2.9	14.0	28.6	48.5	0.0	0.0	0.0
R13	0.8	1.1	1.3	1.6	7.8	16.2	29.7	41.7	0.0	0.0
R14	0.7	0.9	1.1	1.5	7.1	15.3	29.1	44.3	0.0	0.0
R15	0.6	0.8	1.0	1.3	6.6	14.5	28.6	46.6	0.0	0.0
R16	0.5	0.7	0.9	1.2	6.1	13.8	28.1	48.7	0.0	0.0
R17	0.3	0.4	0.4	0.6	3.2	7.4	15.4	28.6	43.6	0.0
R18	0.2	0.3	0.4	0.6	2.9	7.0	14.8	28.4	45.3	0.0
R19	0.2	0.3	0.3	0.5	2.7	6.6	14.3	28.1	47.9	0.0
R20	0.2	0.3	0.3	0.5	2.5	6.1	13.7	27.7	48.8	0.0
R21	0.1	0.1	0.1	0.2	1.2	3.1	7.1	14.7	27.5	45.8
R22	0.1	0.1	0.1	0.2	1.1	2.9	6.8	14.3	27.4	47.0
R23	0.1	0.1	0.1	0.2	1.0	2.7	6.4	13.8	27.3	48.2
R24	0.0	0.1	0.1	0.2	0.9	2.5	6.1	13.4	27.2	49.4
R25	0.0	0.1	0.1	0.2	0.9	2.4	5.9	13.1	26.8	50.6
R26	0.0	0.1	0.1	0.1	0.8	2.2	5.6	12.7	26.6	51.8
R27	0.0	0.1	0.1	0.1	0.7	2.1	5.3	12.2	26.4	53.0
R28	0.0	0.1	0.1	0.1	0.7	1.9	5.0	11.8	26.1	54.2

Table 4: The table shows the percentage probability of FedEx Cup outcomes: $\delta = 16.75$.

More important in determining the outcome is the number of rounds that a player must play to win. This provides a major incentive to the players to improve their ranking during the first three playoff-series events by participating and playing well. It is therefore an important justification for the strong seed-ing approach that we propose.

We now discuss the issues of rest and a reduction in recent match practice that arise when players are required to play various numbers of matches, as in our strongly seeded playoff system. The comparison with the impact of these factors in other sports is problematic. For example, in the National Football League, rest is perceived as an advantage because it provides players with an opportunity to recover from the frequent injuries that occur. Professional golf, by contrast, has a low (although nonzero) injury rate. This means that rest may not be a big factor. The

issue of a reduction in recent match practice may also not be a significant factor for two reasons. First, our proposed playoff design ensures that the players will probably have played competitively in the three previous tournaments. Second, the small field size at the TOUR Championship permits players who are not yet active in the competition to practice on the same course at the end of the day. This is convenient because each player knows in advance when his first match will take place, and he can plan his practice so that he can reach peak form for that match.

Advantages of the Proposed Design

In this section, we discuss several ways in which our proposed FedEx Cup design meets the objectives discussed in the *Outcomes and Commentary* section.

OBJECTIVE 1. The playoffs should build towards a climactic finish, meaning that the FedEx Cup should be decided at the TOUR Championship.

1. Similar to playoff systems in most other major sports, the results of the final event, the TOUR Championship, determine who wins, as well as all the other positions in the final standings. This adds credibility both to the FedEx Cup and to the sport of professional golf.

2. All the marquee name players (i.e., those ranked first through fourth) are guaranteed to reach the Saturday afternoon network television audience. This often fails to happen in weakly seeded match-play tournaments, such as the WGC-Accenture match-play championship. In addition, these four players are each guaranteed at least a tie for fifth place, which is justified by their consistent performance earlier in the season.

3. During the latter stages of the event, the players start even. For example, at midday on Sunday, the two finalists start even. This avoids the anticlimax created in a stroke-play event when one player has a substantial lead on the field after three, or even after two, rounds.

4. Based on our probabilistic analysis, the results of the proposed design provide an appropriate balance between the top-ranked players and those who are ranked lower. The top-ranked player wins about 16.5 percent of the time, and the other players ranked in the top four collectively win 42.6 percent of the

time. This suggests that the final contenders will typically be marquee name players, thus increasing the event's authenticity. Nonetheless, the openness of the event is illustrated by, for example, players ranked 9th through 12th collectively winning more than 10 percent of the time. This adds excitement and uncertainty to the event. It is important to note that these numerical results are not sensitive to the assumption we make about the frequency with which a higher-ranked player beats a lower-ranked player in an individual match.

OBJECTIVE 2. The FedEx Cup competition should culminate in a dramatic way, meaning that a reasonable number of players should be in strong contention to win it at the TOUR Championship.

1. All 28 players competing at the TOUR Championship control their own destiny in the FedEx Cup. For the 28th-ranked player to win, he needs to win all of his eight matches. Although this task is difficult, the numerical results obtained from our probabilistic analysis indicate that it is possible. This meets Objective 2 better than almost any other design can. We anticipate that both the players and the spectators would enjoy this feature of the proposed design.

2. The possibility of a lesser-known player making his way through the field and eventually challenging one of the marquee name players in the later stages of the TOUR Championship adds greatly to the excitement of the event.

OBJECTIVE 3. The TOUR Championship field should consist of the top performers in the regular season in addition to players who have excelled in the playoffs.

1. The 28 players who are invited to compete at the TOUR Championship are determined by their regular-season performance and their performance in the first three playoff-series events.

OBJECTIVE 4. A player's performance during the regular season should impact his finishing position in the FedEx Cup and his resulting prize money.

1. Strong performance during the regular season and the first three playoff-series events greatly improves a player's chance of winning the FedEx Cup or finishing in a high position. For example, a player who ranks 25th–28th after the first three playoff-series

events must win eight matches at the TOUR Championship to win the FedEx Cup. However, a player who ranks first through fourth must win only three matches to do so. This reduces concerns among the top players about having consistently good performance during the season overridden by the results of one playoff event.

2. Players who have a high ranking coming into the TOUR Championship are guaranteed to receive substantial prize money from that event and also from the FedEx Cup, even if they lose the first match they play. For example, a player who ranks 9th–12th cannot finish lower than T13th in the FedEx Cup, and a player who ranks 13th–16th cannot finish lower than T17th. In both cases, the player wins substantial prize money.

Additional Advantages

1. Various stakeholders, such as golf writers, believe that the one-on-one format of match play is much closer to the spirit of a playoff event than stroke play is.

2. The strong seeding system we propose for the TOUR Championship provides substantial incentives for the players to participate and perform well at the first three playoff-series events (see Table 5). Columns 2 through 5 in this table show the rank of the player entering the playoff series, and after the first, second, and third playoff-series event, respectively. Based on the complete results of the playoff-series events, we identify for each player one event (or more for some players) that he could have missed with the least reduction in ranking. The sixth column identifies those events and what that ranking would have been. The last column shows how many extra rounds the player would consequently need to play to win the FedEx Cup under our proposed design, where “*” denotes that a player would have missed the TOUR Championship if he had missed the event indicated.

The results in Table 5 identify several distinctions between the current and proposed designs of the TOUR Championship. Jim Furyk and Zach Johnson would probably be unconcerned about dropping from third and fourth place, respectively, to fifth place under the current design. However, Table 2 shows that this reduces their probability of winning the

Player	Rank before event 1	Rank after event 1	Rank after event 2	Rank after event 3	Events to miss: Rank	Extra rounds
Tiger Woods	1	1	2	1	1, 2:1	0
Steve Stricker	2	2	1	2	1, 3:2	0
Jim Furyk	16	18	18	3	1:5	1
Zach Johnson	3	4	4	4	1:5	1
Heath Slocum	124	3	3	5	3:5	0
Padraig Harrington	66	14	7	6	3:13	2
Sean O'Hair	10	16	16	7	1:7	0
Scott Verplank	61	37	5	8	3:10	1
Kenny Perry	4	5	6	9	3:10	0
Jason Dufner	42	57	9	10	1:10	0
Dustin Johnson	19	20	10	11	3:17	2
Nick Watney	13	7	11	12	2:17	2
Geoff Ogilvy	8	12	8	13	1:13	0
Phil Mickelson	6	10	12	14	1:17	1
Retief Goosen	11	17	17	15	1:15	0
Marc Leishman	78	93	67	16	1:16	0
Brian Gay	9	8	13	17	2:22	1
Kevin Na	23	22	23	18	2:27	1
David Toms	12	13	19	19	2:24	1
Lucas Glover	5	9	14	20	1, 2, 3:24	1
Y. E. Yang	7	6	15	21	2:21	0
Ernie Els	47	11	20	22	2:22	0
Hunter Mahan	14	15	21	23	2:25	0
Angel Cabrera	27	28	22	24	3:29T	*
Steve Marino	30	23	26	25	2:32	*
Stewart Cink	15	19	25	26	2:26	0
Mike Weir	26	29	27	27	2:32	*
Luke Donald	24	25	32	28	2:32	*

Table 5: The table illustrates the progress of the top 28 TOUR Championship players in the 2009 playoff series.

FedEx Cup from 14.1 percent and 13.2 percent, respectively, to 7.1 percent, because of the need to play an extra round. Similarly, the win probabilities of Dustin Johnson and Nick Watney would decrease from 2.4 percent and 2.2 percent, respectively, to 0.4 percent, because of the need to play two extra rounds if they miss their indicated playoff-series event. These examples show that the consequences of a small drop in the rankings can be much greater under our proposed design than under the current design. Hence, the incentive to play in every playoff-series event is increased.

3. Many spectators, both at the course and watching on television, can enjoy knowing that their favorite player, even if not necessarily among the top players, has a real chance to win.

4. The proposed format is easier for the fans to understand, and therefore more exciting, than the current points system.

Extensions

Various possible extensions or variations of our proposed design could be used to add further excitement to the event.

1. If the players (other than the four semifinalists) are willing to play two additional matches after they lose, then it would be possible to rank all 28 players. This proposal would also provide more matches for spectators to watch. For example, there would be eight matches (instead of four) on Saturday afternoon, six matches (instead of two) on Sunday morning, and four matches (instead of two) on Sunday afternoon. Thus, the total number of matches would increase from 28 to 52.

2. Alternatively, the system described above could be used only for some of the leading players, for example, the top eight. In this case, there would be

four matches on Sunday morning and four on Sunday afternoon.

3. Ties, for example, between fifth and eighth place, could be broken by using the total points entering the TOUR Championship.

4. The last one or two places in the TOUR Championship field could be determined by a playoff. For example, this could be used to allow one of the year's major champions who was not in the top 28 players in the points standings before the TOUR Championship to play his way into that event. A similar system is used to determine the 64th and last place in the NCAA Basketball Championship.

Conclusions

In this paper, we propose a new design for the FedEx Cup playoff series on the PGA TOUR. We believe that the results of the FedEx Cup playoff series should be determined by its final event, the TOUR Championship. We also propose redesigning the TOUR Championship as a strongly seeded match-play event with 28 players. To assess the fairness of the proposed design to the players, we perform a probabilistic analysis that determines how a player's rank entering the TOUR Championship influences his finishing position in the FedEx Cup. This analysis confirms that the expected results are fair to players of various ranks. Among the many advantages of the proposed design are that all players have a realistic chance to win, that players with better regular season and earlier playoff performance have a greater chance to win, and that participation of the marquee name players in the closing stages of the event is guaranteed. Because of the inherent excitement and uncertainty of match-play events, we anticipate that our design will provide both an exciting climax to the professional golf season and a highly marketable television product.

We discuss the possible reactions of several stakeholders in professional golf to our proposed system. First, as discussed in the *Outcomes and Commentary* section, golf writers are overwhelmingly in favor of match play. There is a strong consensus that the head-to-head format of match play provides a truer playoff system than conventional stroke play, and is also more consistent with playoff systems in other major

sports. Second, we believe that these opinions reasonably represent the opinions of the on-course and television audiences. Those audiences should enjoy the predictability of seeing the marquee name players in action late in the event, and the unpredictability of each individual match. Third, PGA TOUR administrators are wary of match play, apparently because of the risk of the marquee name players being eliminated early in the competition. However, we believe that our proposed playoff system resolves that concern. Finally, professional golfers, like other athletes, are typically very competitive. Whereas under the current system only 5 of the 30 players at the TOUR Championship control their own destiny, under the proposed system all 28 players do so. We expect that the players would be excited about this.

In conclusion, we believe that our proposed design comes closer than any other design to meeting the objectives outlined by Commissioner Finchem for the FedEx Cup playoff series.

Appendix

Using the definitions in the *Probabilistic Analysis* section, we compute the following probabilities:

$$p_{j_1, j_2, j_3, j_4}(r) = p_{j_1, j_2, j_3, j_4}(r-1)q_{j_1, i_1}q_{j_2, i_2}q_{j_3, i_3}q_{j_4, i_4} \quad (4)$$

$$p_{j_1, j_2, j_3, i_4}(r) = \sum_{j_4=j_4+1}^{j_3-1} p_{j_1, j_2, j_3, j_4}(r-1) \cdot q_{j_1, i_1}q_{j_2, i_2}q_{j_3, i_3}q_{i_4, j_4}, \quad (5)$$

$$p_{j_1, j_2, j_4, i_3}(r) = \sum_{j_3=j_3+1}^{j_2-1} p_{j_1, j_2, j_3, j_4}(r-1) \cdot q_{j_1, i_1}q_{j_2, i_2}q_{i_3, j_3}q_{j_4, i_4}, \quad (6)$$

$$p_{j_1, j_3, j_4, i_2}(r) = \sum_{j_2=j_2+1}^{j_1-1} p_{j_1, j_2, j_3, j_4}(r-1) \cdot q_{j_1, i_1}q_{i_2, j_2}q_{j_3, i_3}q_{j_4, i_4}, \quad (7)$$

$$p_{j_2, j_3, j_4, i_1}(r) = \sum_{j_1=j_1+1}^{28} p_{j_1, j_2, j_3, j_4}(r-1) \cdot q_{i_1, j_1}q_{j_2, i_2}q_{j_3, i_3}q_{j_4, i_4}, \quad (8)$$

$$p_{j_1, j_2, i_3, i_4}(r) = \sum_{j_3=j_3+2}^{j_2-1} \sum_{j_4=j_4+1}^{j_3-1} p_{j_1, j_2, j_3, j_4}(r-1) \cdot q_{j_1, i_1}q_{j_2, i_2}q_{i_3, j_3}q_{i_4, j_4}, \quad (9)$$

$$p_{j_1, j_3, i_2, i_4}(r) = \sum_{j_2=j_2+1}^{j_1-1} \sum_{j_4=j_4+1}^{j_3-1} p_{j_1, j_2, j_3, j_4}(r-1) \cdot q_{j_1, i_1}q_{i_2, j_2}q_{j_3, i_3}q_{i_4, j_4}, \quad (10)$$

$$p_{j_2, j_3, i_1, i_4}(r) = \sum_{j_1=j_2+1}^{28} \sum_{j_4=i_4+1}^{j_3-1} p_{j_1, j_2, j_3, j_4}(r-1) \cdot q_{i_1, j_1} q_{j_2, i_2} q_{j_3, i_3} q_{i_4, j_4}, \quad (11)$$

$$p_{j_1, j_4, i_2, i_3}(r) = \sum_{j_2=j_4+2}^{j_1-1} \sum_{j_3=j_4+1}^{j_2-1} p_{j_1, j_2, j_3, j_4}(r-1) \cdot q_{j_1, i_1} q_{i_2, j_2} q_{i_3, j_3} q_{j_4, i_4}, \quad (12)$$

$$p_{j_2, j_4, i_1, i_3}(r) = \sum_{j_1=j_2+1}^{28} \sum_{j_3=j_4+1}^{j_2-1} p_{j_1, j_2, j_3, j_4}(r-1) \cdot q_{i_1, j_1} q_{j_2, i_2} q_{i_3, j_3} q_{j_4, i_4}, \quad (13)$$

$$p_{j_3, j_4, i_1, i_2}(r) = \sum_{j_1=j_3+2}^{28} \sum_{j_2=j_3+1}^{j_1-1} p_{j_1, j_2, j_3, j_4}(r-1) \cdot q_{i_1, j_1} q_{i_2, j_2} q_{j_3, i_3} q_{j_4, i_4}, \quad (14)$$

$$p_{j_1, i_2, i_3, i_4}(r) = \sum_{j_2=i_4+3}^{j_1-1} \sum_{j_3=i_4+2}^{j_2-1} \sum_{j_4=i_4+1}^{j_3-1} p_{j_1, j_2, j_3, j_4}(r-1) \cdot q_{j_1, i_1} q_{i_2, j_2} q_{i_3, j_3} q_{i_4, j_4}, \quad (15)$$

$$p_{j_2, i_1, i_3, i_4}(r) = \sum_{j_1=j_2+1}^{28} \sum_{j_3=i_4+2}^{j_2-1} \sum_{j_4=i_4+1}^{j_3-1} p_{j_1, j_2, j_3, j_4}(r-1) \cdot q_{i_1, j_1} q_{j_2, i_2} q_{i_3, j_3} q_{i_4, j_4}, \quad (16)$$

$$p_{j_3, i_1, i_2, i_4}(r) = \sum_{j_1=j_3+2}^{28} \sum_{j_2=j_3+1}^{j_1-1} \sum_{j_4=i_4+1}^{j_2-1} p_{j_1, j_2, j_3, j_4}(r-1) \cdot q_{i_1, j_1} q_{i_2, j_2} q_{j_3, i_3} q_{i_4, j_4}, \quad (17)$$

$$p_{j_4, i_1, i_2, i_3}(r) = \sum_{j_1=j_4+3}^{28} \sum_{j_2=j_4+2}^{j_1-1} \sum_{j_3=j_4+1}^{j_2-1} p_{j_1, j_2, j_3, j_4}(r-1) \cdot q_{i_1, j_1} q_{i_2, j_2} q_{i_3, j_3} q_{i_4, j_4}, \quad (18)$$

$$p_{i_1, i_2, i_3, i_4}(r) = \sum_{j_1=i_4+4}^{28} \sum_{j_2=i_4+3}^{j_1-1} \sum_{j_3=i_4+2}^{j_2-1} \sum_{j_4=i_4+1}^{j_3-1} p_{j_1, j_2, j_3, j_4}(r-1) \cdot q_{i_1, j_1} q_{i_2, j_2} q_{i_3, j_3} q_{i_4, j_4}. \quad (19)$$

In a similar way, the probabilities $p_{j_1, j_2, j_3, j_4}(7)$ are evaluated recursively from the probabilities of progressing from round 6 using

$$p_{j_1, j_2, j_3, j_4}(7) = \begin{cases} p_{j_1, j_2, j_3, j_4}(6) q_{j_1, i_4} q_{j_2, j_3} & \text{for } j_2 > j_3, \\ p_{j_1, j_3, j_2, j_4}(6) q_{j_1, i_4} q_{j_2, j_3} & \text{for } j_1 > j_3 > j_2 > j_4, \\ p_{j_3, j_1, j_4, j_2}(6) q_{j_1, i_4} q_{j_2, j_3} & \text{for } j_3 > j_1 > j_4 > j_2, \\ p_{j_3, j_4, j_1, j_2}(6) q_{j_1, i_4} q_{j_2, j_3} & \text{for } j_4 > j_1. \end{cases} \quad (20)$$

Finally, we evaluate the probabilities of players j_1, j_2, j_3 , and j_4 occupying the first four positions, $p_{j_1, j_2, j_3, j_4}(8)$, based on

the probabilities of results achieved in the semifinals using

$$p_{j_1, j_2, j_3, j_4}(8) = \begin{cases} p_{j_1, j_2, j_3, j_4}(7) q_{j_1, j_2} q_{j_3, j_4} & \text{for } j_1 > j_2, j_3 > j_4, \\ p_{j_1, j_2, j_4, j_3}(7) q_{j_1, j_2} q_{j_3, j_4} & \text{for } j_1 > j_2, j_4 > j_3, \\ p_{j_2, j_1, j_3, j_4}(7) q_{j_1, j_2} q_{j_3, j_4} & \text{for } j_2 > j_1, j_3 > j_4, \\ p_{j_2, j_1, j_4, j_3}(7) q_{j_1, j_2} q_{j_3, j_4} & \text{for } j_2 > j_1, j_4 > j_3. \end{cases} \quad (21)$$

As we show in the *Probabilistic Analysis* section, the probability of player j winning the TOUR Championship is determined from Equation (2). The probabilities of player j finishing second, third, or fourth are evaluated analogously. It is also straightforward to use the probabilities computed in Equations (1) and (4) through (19) to evaluate the probability of a player finishing in a tie for position 5, 9, 13, 17, 21, or 25.

As a first example, we consider the probability of player k finishing in a tie for fifth place. The evaluation of this probability depends on whether k is one of the top-four-ranked players who then is eliminated in his first match in round 6.

The probability of player i_k , where $i_k \in \{1, 2, 3, 4\}$, finishing in a tie for fifth place is given by

$$\sum_{j_1=8}^{28} \sum_{j_2=7}^{j_1-1} \sum_{j_3=6}^{j_2-1} \sum_{j_4=5}^{j_3-1} p_{j_1, j_2, j_3, j_4}(5) q_{j_k, i_k}.$$

Alternatively, the probability of player k , where $k \notin \{1, 2, 3, 4\}$, finishing in a tie for fifth place is given by

$$\begin{aligned} & \sum_{j_2=7}^{k-1} \sum_{j_3=6}^{j_2-1} \sum_{j_4=5}^{j_3-1} p_{k, j_2, j_3, j_4}(5) q_{j_1, k} + \sum_{j_1=k+1}^{28} \sum_{j_3=6}^{k-1} \sum_{j_4=5}^{j_3-1} p_{j_1, k, j_3, j_4}(5) q_{2, k} \\ & + \sum_{j_1=k+2}^{28} \sum_{j_2=k+1}^{j_1-1} \sum_{j_4=5}^{k-1} p_{j_1, j_2, k, j_4}(5) q_{3, k} \\ & + \sum_{j_1=k+3}^{28} \sum_{j_2=k+2}^{j_1-1} \sum_{j_3=k+1}^{j_2-1} p_{j_1, j_2, j_3, k}(5) q_{4, k}. \end{aligned} \quad (22)$$

Note that in Equation (22) we adopt the standard convention that a summation is equal to zero if its lower limit exceeds its upper limit. Thus, for $k = 5$ only the last term of Equation (22) is nonzero, whereas for $k = 28$ only the first term is nonzero.

As a second example, we provide a detailed computation of the probability that the player ranked 25th finishes in a tie for position 21. Observe that this player beats the player ranked 24th in round 1, and then loses in round 2. The winning opponent in round 2 depends on the relative ranking of the four players who advance from round 1. There are four possibilities. We consider them in turn.

1. The player ranked 25th is the *bottom-ranked* player advancing from round 1. Here, the players advancing from round 1 have ranks (25, 23, 22, 21). This occurs with

probability $q_{25,24}q_{23,26}q_{22,27}q_{21,28}$. In this case, the round 2 opponent has rank 17 and wins with probability $q_{17,25}$.

2. The player ranked 25th is the *third-ranked* player advancing from round 1. There are three alternatives for the players advancing from round 1:

(a) ranks (28, 25, 23, 22) with probability $q_{28,21} \cdot q_{25,24}q_{23,26}q_{22,27}$;

(b) ranks (27, 25, 23, 21) with probability $q_{27,22} \cdot q_{25,24}q_{23,26}q_{21,28}$; and

(c) ranks (26, 25, 22, 21) with probability $q_{26,23} \cdot q_{25,24}q_{22,27}q_{21,28}$.

In each case, the round 2 opponent has rank 18 and wins with probability $q_{18,25}$.

3. The player ranked 25th is the *second-ranked* player advancing from round 1. There are three alternatives for the players advancing from round 1:

(a) ranks (28, 27, 25, 23) with probability $q_{28,21} \cdot q_{27,22}q_{25,24}q_{23,26}$;

(b) ranks (28, 26, 25, 22) with probability $q_{28,21} \cdot q_{26,23}q_{25,24}q_{22,27}$; and

(c) ranks (27, 26, 25, 21) with probability $q_{27,22} \cdot q_{26,23}q_{25,24}q_{21,28}$.

In each case, the round 2 opponent has rank 19 and wins with probability $q_{19,25}$.

4. The player ranked 25th is the *top-ranked* player advancing from round 1. The players advancing from round 1 have ranks (28, 27, 26, 25). This occurs with probability $q_{28,21}q_{27,22}q_{26,23}q_{25,24}$. In this case, the round 2 opponent has rank 20 and wins with probability $q_{20,25}$.

Combining these four possibilities, the probability of the player ranked 25th finishing in a tie for position 21 is obtained by adapting Equation (22) to give

$$\begin{aligned} & \sum_{j_2=23}^{24} \sum_{j_3=22}^{j_2-1} \sum_{j_4=21}^{j_3-1} p_{25,j_2,j_3,j_4} (1) q_{17,25} + \sum_{j_1=26}^{28} \sum_{j_3=22}^{24} \sum_{j_4=21}^{j_3-1} p_{j_1,25,j_3,j_4} (1) q_{18,25} \\ & + \sum_{j_1=27}^{28} \sum_{j_2=26}^{j_1-1} \sum_{j_4=21}^{24} p_{j_1,j_2,25,j_4} (1) q_{19,25} + \sum_{j_1=28}^{28} \sum_{j_2=27}^{j_1-1} \sum_{j_3=26}^{j_2-1} p_{j_1,j_2,j_3,25} (1) q_{20,25} \\ & = q_{25,24}q_{23,26}q_{22,27}q_{21,28}q_{17,25} + (q_{28,21}q_{25,24}q_{23,26}q_{22,27} \\ & + q_{27,22}q_{25,24}q_{23,26}q_{21,28} + q_{26,23}q_{25,24}q_{22,27}q_{21,28})q_{18,25} \\ & + (q_{28,21}q_{27,22}q_{25,24}q_{23,26} + q_{28,21}q_{26,23}q_{25,24}q_{22,27} \\ & + q_{27,22}q_{26,23}q_{25,24}q_{21,28})q_{19,25} + q_{28,21}q_{27,22}q_{26,23}q_{25,24}q_{20,25}, \end{aligned}$$

from (1), (4), (6), (7), (8), (12), (13), (14), and (18).

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